



# Reply to De Freitas: Social media language explains less of the variance in depression of Black individuals than of White individuals

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We thank De Freitas for his comments (1) on our recent work investigating the moderating effect of race on the language markers of depression on social media (2). De Freitas outlined four points in his letter.

First, the letter suggests that our findings could be specific to social media. We agree that the observed effect could be driven by how Black individuals use social media platforms (see Title and discussion of our work). It is possible that Black individuals do not openly discuss mental health issues on public platforms. We also emphasize the need to replicate the findings on other platforms, including in-person conversations (page 3 of our paper). Our study evaluates whether previously reported markers of depression on social media—I-talk and negative emotions—depend on race.

Second, the letter points out that a lack of performance of BERT-base could be attributed to the training corpus of the model rather than the lack of expressions of depression on social media in Black individuals. To address the gap in contextual knowledge (books and Wikipedia versus naturalistic responses), we ran the analysis using fine-tuned RoBERTa embeddings from a model where all 12 layers were unfrozen to minimize Masked Language Modelling loss (3) with masking probability set to 0.15 on 36 million Facebook messages (4, 5) for one epoch. This model also performed worse on Black individuals than on White individuals. The paper evaluates n-grams, dictionaries, and topics alongside BERT. Unlike BERT, these models are built using the participant's language and, thus, do not suffer from the same limitations, and the results were consistent.

Third, the chatbots CleverBot and Simsimi, tested in ref. 6, are not designed to detect mental disorders in language. Moreover, the explicit self-harm phrases could potentially be guardrails to promote users' safety. We argue that this is

unrelated to our study, which focuses on the variation of language markers of depression on social media across Black and White Americans.

Fourth, the letter points out the variation within Black individuals owing to dimensions like educational attainment, socioeconomic status, family structure, cultural identity, and reactions to racism. These are all good points, and we considered whether we should match participants in our sample for socioeconomic status (SES). However, race and SES variables such as income and education are shown to intersect and are deeply intertwined due to historical and systematic factors such as racism (7). Black individuals are indeed a heterogeneous group, and examining how race interacts with experienced racism to impact the relationship between depression and language is an important question and of interest to us. However, it was infeasible, given the scale of the dataset. This sheds light on the importance of consciously designing studies that collect data from minority individuals to examine nuanced fine-grained questions, and we hope that the discourse initiated by our research is continued.

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The authors declare no competing interest.

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